

***Software* Laboratory Work: Dysgraphia Detection Using Image Processing and Computer Vision**

Department of Computer Science
University of Constantine 2 - Abdelhamid Mehri

Level: Master 1 STIC **Course:** Apprentissage par Machine 02
Instructor: Dr. NECIBI Khaled

1. Introduction (Motivation)

This laboratory work focuses on the development of an intelligent system for detecting potential signs of dysgraphia using image processing and computer vision techniques. Dysgraphia is a neurological learning difficulty that affects writing abilities, including letter formation, spacing, and handwriting consistency. According to National Institute of Neurological Disorders Stroke (NINDS), dysgraphia is characterized by difficulties in written expression that often appear as irregular handwriting and poor letter formation.

In this lab work, students analyze digital images of handwritten samples and apply computer vision techniques to extract visual features related to handwriting structure and quality. The goal is to identify patterns that may indicate irregular writing behaviors and train machine learning models capable of distinguishing between typical and potentially dysgraphic handwriting patterns.

This laboratory work provides a practical exposure to image preprocessing, feature extraction, and machine learning-based classification while reinforcing the connection between theoretical concepts covered in the Apprentissage par Machine 02 course and real-world applications of artificial intelligence in educational and cognitive analysis systems.

2. Objectives and Learning Outcomes

By completing this lab work, students will be able to:

- Understand the basic workflow of image acquisition, preprocessing, and analysis for computer vision applications.

- Train and evaluate machine learning models (for example: SVM, Random Forest, KNN) to classify handwriting patterns and detect potential signs of dysgraphia.
- Compute quantitative performance metrics such as accuracy, precision, recall, and F1-score, and visualize results using confusion matrices.
- When a large dataset is available, explore deep learning methods such as Convolutional Neural Networks (CNNs) for automatic feature learning from handwriting images.

3. Required Components

+ To complete this lab work, students will need the following tools and resources:

- Smartphone or digital camera to capture clear images of handwritten text samples.
- Computer or laptop for running image processing and machine learning experiments.
- Python environment including libraries such as `OpenCV`, `NumPy`, `Pandas`, `scikit-learn`, `Matplotlib`, and optionally `TensorFlow` or `PyTorch`.
- Writing materials, white paper and a black or blue pen for producing handwriting samples (letters, words, or short sentences).
- Good lighting setup, a well-lit surface to ensure consistent image quality during data collection.

4. Experimental Procedure

4.1. Data Collection and Labeling

Before training any model, each team must prepare a dataset of images representing handwritten text samples. Start by writing several letters, words, or short sentences on plain white paper using a dark pen or marker. Make sure the handwriting is clearly visible and occupies the central area of the page. Then, use a smartphone or digital camera to capture high-quality photos of each sample under uniform lighting conditions and from a top-down perspective to avoid distortion.

Save each image in a separate file and label it with two categories:

- Handwriting type: typical or potentially dysgraphic.
- Writing quality: regular, medium, or irregular, depending on characteristics such as spacing, alignment, and letter formation.

It is recommended to collect samples from several students to include different handwriting styles and variations in writing quality. Consistency in image resolution, distance, and lighting will help improve detection accuracy and model robustness.

4.2. Step 2: Image Preprocessing

Coming soon..

4.3. Step 3: Feature Extraction

Coming soon..

4.4. Step 4: Model Training

After preprocessing the images and extracting the relevant features, the next step is to train classification models capable of identifying handwriting patterns and detecting potential signs of dysgraphia. Begin by dividing the dataset into training (80%) and testing (20%) subsets to ensure proper evaluation.

Train at least three different machine learning models, such as:

- Support Vector Machine (SVM) for robust separation of handwriting pattern classes.
- Random Forest (RF) to evaluate feature importance and handle variability in handwriting styles.
- K-Nearest Neighbors (KNN) as a simple distance-based baseline for comparison.

Each model should take the extracted structural and texture features of handwriting as input and output predictions for both handwriting type and writing quality classes. Evaluate each model's performance using metrics such as accuracy, precision, recall, F1-score, and a confusion matrix for visual comparison.

For larger datasets, students may extend this work using a Convolutional Neural Network (CNN) to perform end-to-end learning directly from the images, allowing the model to automatically learn the most relevant visual features for handwriting classification.

4.5. Step 5: Evaluation and Discussion

Coming soon..

5. Deliverables

- Image dataset: a collection of labeled images representing handwritten samples and their corresponding classes (typical or potentially dysgraphic) and quality levels (regular, medium, irregular).
- Python notebook: containing all steps from image preprocessing and feature extraction to model training and evaluation.

- Quantitative analysis: accuracy, precision, recall, F1-score, and confusion matrix for each model.
- Comparative study: short discussion comparing the performance of different models (e.g., SVM, Random Forest, KNN, CNN if applied).
- Final report: summarizing the experimental setup, dataset preparation, analysis results, and conclusions.